



Institute of Technology of Cambodia
Department of Electrical and Energy Engineering



Computer Programming Mini Project:

Zener Diode Voltage Regulator Calculator.

Lab Report by **I3-GEE-C1, Team 4.**

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|-------------------------|----------------------|
| 1. TANG MENGSONG | ID: e20220271 |
| 2. THAI TAIWAN | ID: e20220413 |
| 3. THON SAKSAMBO | ID: e20240038 |
| 4. TANN MENGHAK | ID: e20221590 |
| 5. TAING HYMENG | ID: e20220644 |

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Objective: Calculate the key parameters of a Zener diode voltage regulator circuit, including output voltage, load current, series resistor current, and power dissipated by the Zener diodes.

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1 Introduction

1.1 Background of the Problem

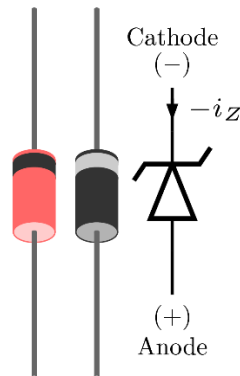


Figure 1.1 Zener Diode ^[1]

There is a need for stable and reliable power supply, voltage source for proper functioning of some electrical components. Voltage regulation is a method to make sure that the voltage is stable for some loads, and Zener diodes are widely used for this purpose due to their **simplicity** and **effectiveness**. By operating in their reverse breakdown mode, Zener diodes maintain a constant output voltage, even if there's variations in the input voltage or load conditions.

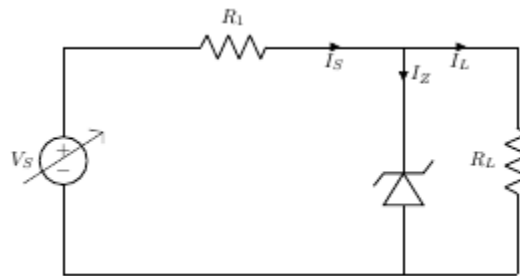


Figure 1.2 Zener diode line regulation ^[1]

A voltage regulator is an electronic circuit that provides a stable DC voltage independent of the load current, temperature and AC line voltage variations. A Zener diode of break down voltage V_Z is reverse connected to an input voltage source V_I across a load resistance R_L and a series resistor R_S . The voltage across the zener will remain steady at its break down voltage V_Z for all the values of zener current I_Z as long as the current remains in the break down region. Hence a regulated DC output voltage $V_0 = V_Z$ is obtained across R_L , whenever the input voltage remains within a minimum and maximum voltage. Basically there are two type of regulations such as: Line Regulation: In this type of regulation, series resistance and load resistance are fixed, only input voltage is changing. Output voltage remains the same as long as the input voltage is maintained above a minimum value. Load Regulation: In this type of regulation, input voltage is fixed and the load resistance is varying. Output volt remains same, as long as the load resistance is maintained above a minimum value. ^[1]

1.2 Rationale for Selecting the Problem

Voltage regulation using Zener diodes is fundamental in electronics. By selecting this problem, the project can achieve:

1. Demonstration of application of Zener diodes in practical scenarios.
2. Can make users simulate and analyze different conditions, such as supply voltage, series resistance, and load resistance.
3. Narrow the gap between theoretical understanding and practical circuit design through program simulation and calculations.

2 Objective of Mini Project

2.1 Objective

Calculate the key parameters of a Zener diode voltage regulator circuit, including output voltage, load current, series resistor current, and power dissipated by the Zener diodes.

2.2 Scope

This project simulates the behavior of a Zener diode voltage regulator using a C program. It calculates key circuit parameters such as **output voltage**, **load current**, **series resistor current**, and **power dissipation**. The program makes sure that the user interface is friendly by putting input validation, error handling, and the ability to perform multiple calculations in one session. While focused on simulation, the project also excludes physical implementation and real-world factors like temperature effects or real component tolerances.

3 Methodology

3.1 Materials

-Hardware used: A running computer with C compiler installed, a text editor or an IDE.

-Software used: Code::Blocks 20.03, MinGW, GNU GCC.

3.2 Mini Project Algorithms

Algorithm Steps:

1. Start the program.
2. Declare the function **calculateZenerParameters(Vin, Vz, Rs, RL)** with return type as **void** and its parameters as **double data type**.
3. Declare variables for input like **Vin** (Supply voltage), **Vz** (Zener breakdown voltage), **Rs** (Zener Diode Series Resistance), **RL** (Load resistance), **result** (result from scanf input), **choice** (user choice whether to exit or continue program).
4. Start with a **do** in **do-while** loop to do calculations if the user chooses to continue the program.
5. (In the do section of do-while loop)
 - Prompt the user to enter **Vin** (Supply Voltage) with the message "Enter supply voltage (Vin) in volts:"

- Read the input using scanf() and at the same time check if the input is validated by getting the value that scanf() outputs into variable **result**. If **result** doesn't return 1 or when **Vin < 0** then display an error message and clear the input buffer.
 - Restart the loop for new input.
 - Repeat the same **step 4** for other variables like **Vz, Rs, RL**
6. Calculate Zener Diode parameters by calling the function **calculateZenerParameters (Vin, Vz, Rs, RL)** with the parameters the user just entered above into the function to process.
 7. (Important) Inside the “**calculateZenerParameters (Vin, Vz, Rs, RL)**” function body: Declare the variables **Vout** (load voltage), **Iout** (load current), **Pz** (Power dissipated by Zener Diode), **Irs** (Series resistor current)
 - Check if **Vin < Vz**, if so, display a message that the Zener diode is not in reverse breakdown mode, that the output voltage is equal to the supply voltage, and ensure that the input voltage is greater than Zener's breakdown voltage to make voltage regulation process work. And terminate the program prematurely.
 - If the check is successful, then the function can begin calculating the parameters for the Zener diode.
 - a. Starting with setting the output voltage (**Vout**) equals to the Zener diode's breakdown voltage only if **Vin > Vz**.
 - b. Calculate load current by $I_{out} = \frac{V_{out}}{R_L}$ and store the result in **Iout** variable.
 - c. Calculate the series resistor's current by $I_{rs} = \frac{V_{in} - V_{out}}{R_s}$ and store the result in **Irs** variable
 - d. Calculate power dissipated by the Zener diode by $P_Z = V_{out} \times (I_{rs} - I_{out})$ and store the results in **Pz** variable.
 - Display the results of **Vout, Iout, Irs** (in milliamps), **Pz** (in milliwatts), and return to the main function.
 8. After executing the function, the user gets to continue or terminate the program by pressing **Y** or **y** for yes or anything else for a “**no**”
 - Start by prompting the user to enter a yes or no answer with “Do you want to perform another calculation?”
 - Clear the input buffer and use scanf to get the user's data and store it in the variable **choice**.
 - Using **while** after **do** in do-while loop in from the beginning of the algorithm to determine if the user wants to repeat the program. If the data from the **choice** variable is “**Y**” or “**y**”, continue the loop, if the input is anything else, then display the message "Thank you for using the Zener diode calculator. Goodbye!" and terminate the program.
 9. Stop the program.

3.3 Mini Project Flowchart

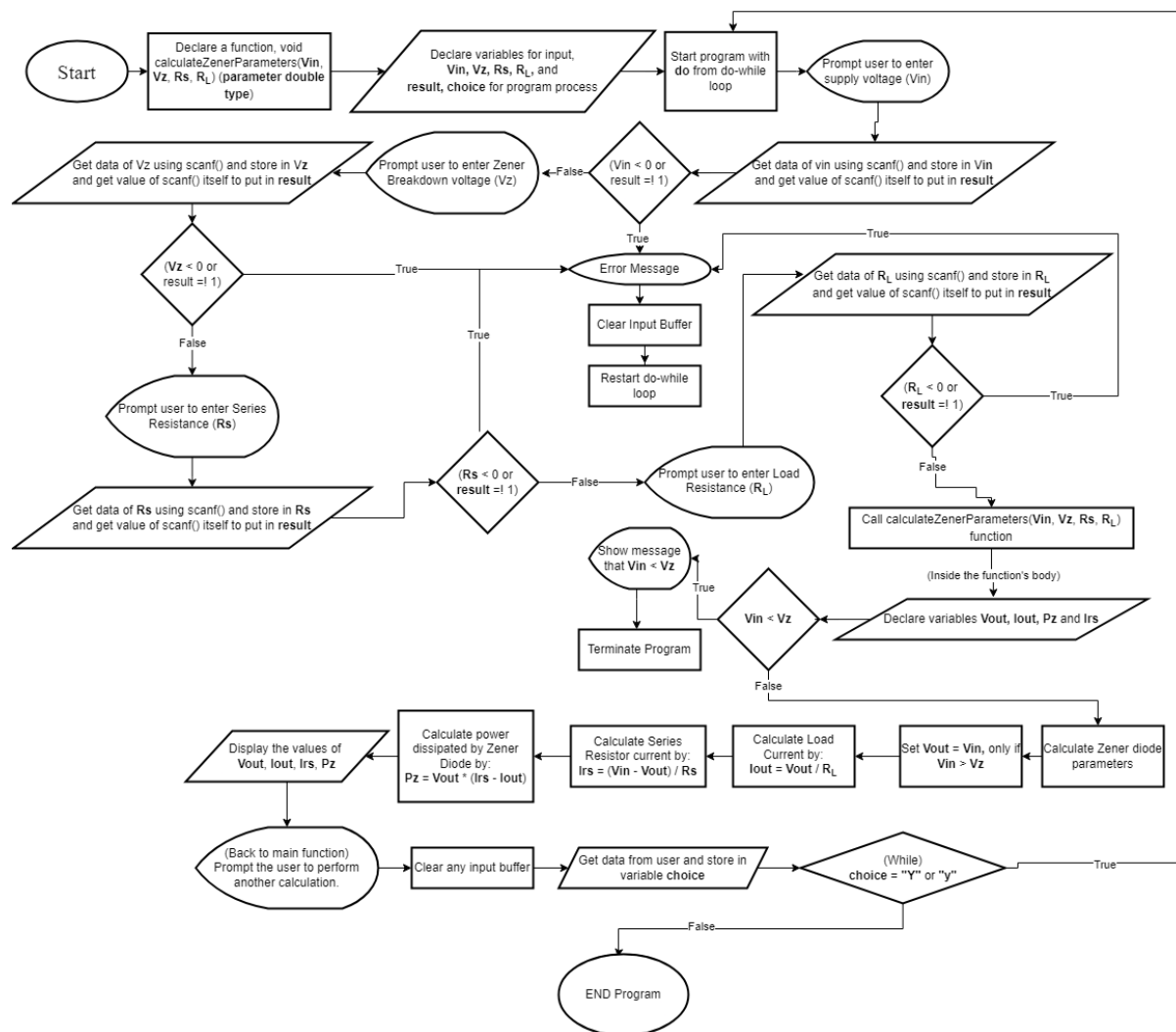


Figure 3.1 Mini Project's flowchart.

3.4 Mini Project C Program to Implement the Solution:

```
#include <stdio.h>
#include <stdlib.h>

void calculateZenerParameters(double Vin, double Vz, double Rs, double RL); // Declare
function first

int main() {
    double Vin, Vz, Rs, RL; // Input variables
    int result; // Variable to store scanf result
    char choice; // Variable for user's choice to continue or exit

    do {
        // Get user inputs and validate them
        printf("Enter supply voltage (Vin) in volts: ");
        result = scanf("%lf", &Vin);
        if (result != 1 || Vin < 0) {
            printf("Invalid input! Supply voltage (Vin) must be a non-negative number.\n");
            while (getchar() != '\n'); // Clear invalid input
            continue; // Restart loop
        }
    }
}
```

```

    printf("Enter Zener breakdown voltage (Vz) in volts: ");
    result = scanf("%lf", &Vz);
    if (result != 1 || Vz < 0) {
        printf("Invalid input! Zener breakdown voltage (Vz) must be a non-negative
number.\n");
        while (getchar() != '\n'); // Clear invalid input
        continue;                // Restart loop
    }

    printf("Enter series resistance (Rs) in ohms: ");
    result = scanf("%lf", &Rs);
    if (result != 1 || Rs < 0) {
        printf("Invalid input! Series resistance (Rs) must be a non-negative
number.\n");
        while (getchar() != '\n'); // Clear invalid input
        continue;                // Restart loop
    }

    printf("Enter load resistance (RL) in ohms: ");
    result = scanf("%lf", &RL);
    if (result != 1 || RL < 0) {
        printf("Invalid input! Load resistance (RL) must be a non-negative number.\n");
        while (getchar() != '\n'); // Clear invalid input
        continue;                // Restart loop
    }

    // Call the function to perform calculations
    calculateZenerParameters(Vin, Vz, Rs, RL);

    // Ask if the user wants to perform another calculation
    printf("\nDo you want to perform another calculation? (y/n): ");
    while (getchar() != '\n'); // Clear input buffer
    scanf("%c", &choice);

} while (choice == 'y' || choice == 'Y'); // Continue if user inputs 'y' or 'Y'

printf("Thank you for using the Zener diode calculator. Goodbye!\n");
return 0;
}

// Function to calculate Zener diode regulator parameters
void calculateZenerParameters(double Vin, double Vz, double Rs, double RL) {
    double Vout, Iout, Pz, Irs;

    if (Vin < Vz) {
        printf("\nThe Zener diode is not in breakdown mode.\n");
        printf("Output Voltage (Vout): %.2f V (equals the supply voltage as the diode is
not regulating)\n", Vin);
        printf("Ensure that the supply voltage (Vin) is greater than the Zener breakdown
voltage (Vz).\n");
        return;
    } else {
        // Calculate output voltage
        Vout = Vz;

        // Calculate load current
        Iout = Vout / RL;

        // Calculate current through series resistor
        Irs = (Vin - Vout) / Rs;

        // Calculate power dissipated by Zener diode
        Pz = Vout * (Irs - Iout);

        // Display results
        printf("\n=== Zener Diode Voltage Regulator Results ===\n");
    }
}

```

```

printf("Output Voltage (Vout): %.2f V\n", Vout);
printf("Load Current (Iout): %.2f mA\n", Iout * 1000);
printf("Current Through Series Resistor (Irs): %.2f mA\n", Irs * 1000);
printf("Power Dissipated by Zener Diode (Pz): %.2f mW\n", Pz * 1000);
}
}

```

3.5 Challenges faced

- **Input Validation:** Making sure that the program can handle invalid inputs, like non-numeric characters or negative numbers. Solutions like clearing the input buffer and checking scanf()'s return value were used to solve this issue.
- **Algorithm Design:** Making a clear, step-by-step algorithm for the calculation loop, including handling cases like $V_{in} < V_z$, is hard.
- **User Experience:** Making sure that the program is user-friendly with prompts, it outputs error messages, and the ability to perform multiple calculations was challenging.

4 Results

- Under normal calculations where no inputs are invalid


```

Enter supply voltage (Vin) in volts: 12
Enter Zener breakdown voltage (Vz) in volts: 10
Enter series resistance (Rs) in ohms: 12000
Enter load resistance (RL) in ohms: 10000

=== Zener Diode Voltage Regulator Results ===
Output Voltage (Vout): 10.00 V
Load Current (Iout): 1.00 mA
Current Through Series Resistor (Irs): 0.17 mA
Power Dissipated by Zener Diode (Pz): -8.33 mW

Do you want to perform another calculation? (y/n): y
Enter supply voltage (Vin) in volts: 15
Enter Zener breakdown voltage (Vz) in volts: 12
Enter series resistance (Rs) in ohms: 10000
Enter load resistance (RL) in ohms: 220000

=== Zener Diode Voltage Regulator Results ===
Output Voltage (Vout): 12.00 V
Load Current (Iout): 0.05 mA
Current Through Series Resistor (Irs): 0.30 mA
Power Dissipated by Zener Diode (Pz): 2.95 mW

Do you want to perform another calculation? (y/n): █

```

Figure 4.1 Normal Operation.

- When one of the input numbers is invalid but the program failed to restart do-while loop

```

Enter supply voltage (Vin) in volts: -12
Invalid input! Supply voltage (Vin) must be a non-negative number.
Thank you for using the Zener diode calculator. Goodbye!

```

Figure 4.2 One of the input numbers is invalid.

- When one of the input data types is invalid

```
Enter supply voltage (Vin) in volts: 24
Enter Zener breakdown voltage (Vz) in volts: 12
Enter series resistance (Rs) in ohms: F
Invalid input! Series resistance (Rs) must be a non-negative number.
Thank you for using the Zener diode calculator. Goodbye!

Process returned 0 (0x0)   execution time : 7.004 s
Press any key to continue.
```

Figure 4.3 One of the input's data types is invalid.

- When $V_{in} < V_z$

```
Enter supply voltage (Vin) in volts: 12
Enter Zener breakdown voltage (Vz) in volts: 24
Enter series resistance (Rs) in ohms: 10000
Enter load resistance (RL) in ohms: 220000

The Zener diode is not in breakdown mode.
Output Voltage (Vout): 12.00 V (equals the supply voltage as the diode is not regulating)
Ensure that the supply voltage (Vin) is greater than the Zener breakdown voltage (Vz).

Do you want to perform another calculation? (y/n):
```

Figure 4.4 Program's output when $V_{in} < V_z$

5 Discussion

- The program can handle invalid inputs and prevent unexpected crashes.
- +The program can show the Zener Diode behavior by:
 - Show proper voltage regulation only when $V_{in} \geq V_z$.
 - Shows correct handling of non-breakdown mode when $V_{in} < V_z$.
- +The program can be modified and scaled to include additional features, such as diode power limits or efficiency analysis.
- +The program cannot restart do-while loop.
- +The program has user-friendly interface with clear prompts and outputs simplifies the simulation of Zener diode voltage regulation.

6 References

- [1] C. B. 3. h. By Teravolt (talk) (Transferred by Nk/Originally uploaded by Teravolt) - I (Teravolt (talk)) created this work entirely by myself. (Originally uploaded on en.wikipedia), "Zener Diode," [Online]. Available: https://en.wikipedia.org/wiki/Zener_diode.
- [2] Virtual Labs, "Zener Diode-Voltage Regulator," [Online]. Available: <https://be-iitkgp.vlabs.ac.in/exp/voltage-regulator/theory.html>.